Fiscal Year:	FY 2019	**Task Last Updated:**	FY 05/15/2020
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Project Title:	Personalizing Prebiotic Therapies for Astronauts' Gut Microbiota		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	27702-3382	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2017 HERO NNJ16ZSA001N-TRIRT. Appendix C: Translational Research Institute for Space Health (TRISH) Research Topics
Start Date:	10/01/2017	**End Date:**	09/30/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Johnson, Sarah Stewart Ph.D. (Georgetown University)		
Grant/Contract No.:	NNX16AO69A-T0106		
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Performance Goal Text:			
	Dietary carbohydrates nourish human gut bacterial communities (microbiota) that resist pathogens, regulate gastrointestinal physiology, and train the immune system. Short-chain fatty acids (SCFAs) like butyrate, which are end products of microbial polysaccharide fermentation, are crucial metabolic precursors and preferred fuel sources for human colonic epithelial cells. Dietary carbohydrates that stimulate the activity of intestinal bacteria (prebiotics) could thus be used to improve astronaut health in spaceflight. Nevertheless, the effects of prebiotic treatments on gut microbiota are known to vary substantially between individuals, likely due to inter-individual microbiota heterogeneity. Practices for how prebiotic therapies should account for microbiota differences remain undiscovered. Without such frameworks, the therapeutic potential of prebiotic therapy for astronauts will remain unrealized. Our long-term research goal is to establish rigorous methods for personalizing prebiotic therapies. Our objective here is		

Task Description:	to develop prebiotic personalization tools that could be applied towards astronauts. This research emerges, in part, from our pioneering studies tracking nutrient shifts in humans on a daily basis for weeks and even up to a year. These studies show that the effects of altering human carbohydrate intake depend on subjects' gut microbiota. Still, fundamental challenges confront efforts to predict prebiotic interactions with microbial communities, including the wide diversity of prebiotics and gut bacterial species, as well as logistical complexities of carrying out human microbiota interventions. To address these challenges, we have developed innovative new tools: a high-throughput microfluidic technique for creating and assaying millions of individual bacterial cultures and, an artificial human intestine that we can sample and manipulate with arbitrary frequency. Here, we will combine these new methods to develop a prebiotic personalization pipeline that can be applied to astronauts. AIM 1: Develop microfluidic culture techniques to measure how different carbohydrate compounds affect the growth and metabolism of the hundreds of distinct gut bacterial species present in microbiota samples. By measuring which bacterial species are stimulated by different carbohydrates, we can individualize prebiotic regimens to maximize butyrate production in a human gut. AIM 2: Validate personalized prebiotic regimens increase butyrate production. We will use our artificial intestine to carry out prebiotic trials on human gut microbial communities with carbohydrates selected by our microfluidic assay in Aim 1. The effects of these regimens on microbiota butyrate output will be compared to non-personalized, "one-size-fits-all" prebiotic treatments. IMPACT: The proposed aims will establish a platform for individualizing prebiotic treatments that could be used to enhance gut bacterial metabolism in astronauts. Customizing prebiotic treatments would also minimize the amount of unused dietary carbohydrates ingested by astronauts, reducing spaceflight payloads. Lastly, by using our artificial intestine to create ex vivo models of astronauts' gut microbiota, we ultimately will be positioned to safely test a variety of astronaut microbiota interventions on Earth, even when these individuals are in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	FINAL REPORTING SUBMITTED DECEMBER 2019 Benefits of this project include: • Development of two separate assays for measuring the potential benefits of over-the-counter prebiotic treatments to human gut microbiota. These assays are applicable to both astronauts as well as the general public. Our assays have also suggested the new biological insight that the ability to degrade multiple dietary polysaccharides has influenced human gut bacterial evolution. • Refinement of an artificial gut model of prebiotic treatment. This experimental model reduces the risks that astronauts need to face when testing new prebiotic treatments. Experiments using this model have already motivated our creation of new statistical models of microbiome dynamics, as well as yielded the insight that prebiotic treatments need to last for multiple several days to have maximum effect. REPORTING SUBMITTED JANUARY 2019 Benefits of this project include: a) Development of two separate assays for measuring the potential benefits of over-the-counter prebiotic treatments to human gut microbiota. These assays are applicable to both astronauts as well as the general public. b) Refinement of an artificial gut model of prebiotic treatment. This experimental model reduces the risks that astronauts need to face when testing new prebiotic treatments. Experiments using this model have already provided the new insight that prebiotic treatments need to last for at least several days to have maximum effect.
Task Progress:	FINAL REPORTING SUBMITTED DECEMBER 2019 The overall goal of this project is to create and validate approaches for personalizing prebiotic treatments to astronauts' gut microbiota. This project pursues that goal through two objectives: 1) Developing and testing in vitro assays for personalizing prebiotic treatments; 2) Validating that assay using both an artificial gut, as well as real-world human samples. The key findings to date of this research are as follows: • We have completed initial development of our microfluidic assays. Findings from this development suggest that most people carry microbes capable of digesting a variety of over-the-counter prebiotics. However, there is individual variation in the total abundance of these microbes and how much they grow on different prebiotics. • In addition, we have also developed a highly efficient second assay for testing individual prebiotic response; this assay yields more direct measurements of gut bacterial metabolism of prebiotics, while being roughly an order of magnitude higher throughput than our microfluidic assay. • Our initial artificial gut experiments show a lag effect where at least two days of prebiotic treatment are necessary for maximal gut bacterial metabolism of prebiotics. * Our preliminary human cohort findings also suggest that a common dietary fiber supplement can elevate levels of beneficial short-chain fatty acids in the human gut. Together, these interim findings support both the need and feasibility of assays to customize prebiotic treatments for astronauts. Our findings also suggest that astronauts will need to consume prebiotics for at least several days before experiencing their maximum benefit. REPORTING SUBMITTED JANUARY 2019 The overall goal of this project is to create and validate approaches for personalizing prebiotic treatments to astronauts' gut microbiota. This project pursues that goal through two objectives: 1) Develop and test a microfluidics-based assay for personalizing prebiotic treatments; 2) Validate that assay using an artificial human gut model. The key findings to date of this research are as follows: a) We have completed initial development of our microfluidic assays. Findings from this development suggest that most people carry microbes capable of digesting a variety of over-the-counter prebiotics. However, there is individual variation in the total abundance of these microbes and how much they grow on different prebiotics. b) In addition, we have also developed a highly efficient second assay for testing individual prebiotic response; this assay yields more direct measurements of gut bacterial metabolism of prebiotics, while being roughly an order of magnitude higher throughput than our microfluidic assay. c) Our initial artificial gut experiments show a "lag effect" where at least two days of prebiotic treatment are necessary for maximal gut bacterial

	metabolism of prebiotics. Together, these interim findings support both the need and feasibility of assays to customize prebiotic treatments for astronauts. Our findings also suggest that astronauts will need to consume prebiotics for at least several days before experiencing their maximum benefit.
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